

भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 360: Machine Learning Lab
Practice Assignment 9

Instructions: This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1. An e-commerce company wants to predict whether a customer will purchase a product based on browsing behaviour and customer characteristics, based on the following attributes with the given values:

   - Age group: Young, middle, senior
   - Income: High, medium, low
   - : Browsing time:long, short
   - previous purchase: Yes,no
   - Purchase (Y/ target): Yes,no

   (a) Implement a Naive Bayes classifier (without using in-built python packages) to predict the target(Purchase) for the test sample: (Age group=middle, Income= High, browsing time=long, previous purchase= Yes).

   (b) Evaluate the deigned classifier using a confusion matrix and report the accuracy, class-wise accuracy, precision and recall.

2. Create a synthetic dataset for a healthcare organization that wants to classify individuals into diabetic and non-diabetic categories based on the following continuous-valued features as below:

   - Glucose level
   - BMI
   - Age

   (a) Implement Gaussian Naive Bayes classifier, (without using in-built python packages) to predict the target(diabetic:yes,no) for the test sample: (Glucose level-142, BMI= 29, Age=42).

   (b) Evaluate the deigned classifier using a confusion matrix and report the accuracy, class-wise accuracy, precision and recall

   (c) Compare the predictions and class conditional probabilities obtained from both implementations (2(a), 3(a)) and report your findings/ observations.

3. Consider the **Diabetes dataset** for classifying patients to be Diabetic, or Non-Diabetic, based on the given predictors.

   (a) Read the dataset.

(b) Implement the ID3 decision tree algorithm using the followings:

1. Information gain to select the best splitting attribute.
2. Thresholding to convert continuous attributes-values to categorical values.

Stopping conditions:

- Pure subsets are obtained (i.e., every observation in the current subset belongs to the same target class).
- Attribute Exhaustion (i.e., no more unused attributes left to split the data).

(c) Evaluate the designed classifier using 5-fold cross-validation. Report the results in terms accuracy, class-wise accuracy, precision and recall.